Computer Vision

Instructor: Luigi Cinque
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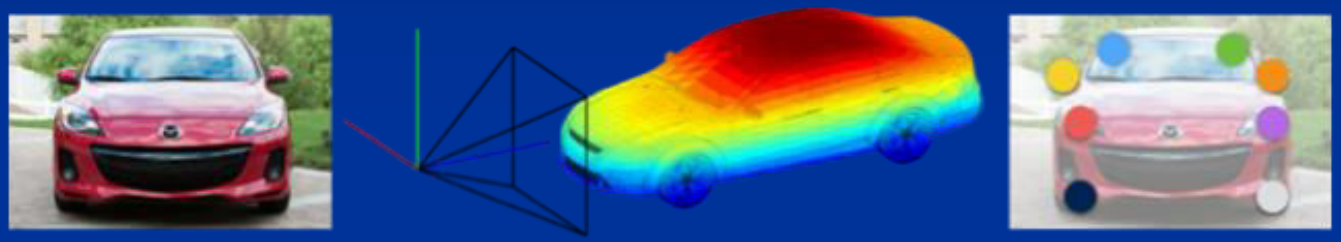
Estimating pose and locating keypoints

Viewpoints and Keypoints

- Shubham Tulsiani, Jitendra Malik



We characterize the problem of pose estimation for rigid objects in terms of determining viewpoint to explain coarse pose and keypoint prediction to capture the finer details. We address both these tasks in two different settings - the constrained setting with known bounding boxes and the more challenging detection setting where the aim is to simultaneously detect and correctly estimate pose of objects. We present Convolutional Neural Network based architectures for these and demonstrate that leveraging viewpoint estimates can substantially improve local appearance based keypoint predictions. In addition to achieving significant improvements over state-of-the-art in the above tasks, we analyze the error modes and effect of object characteristics on performance to guide future efforts towards this goal.

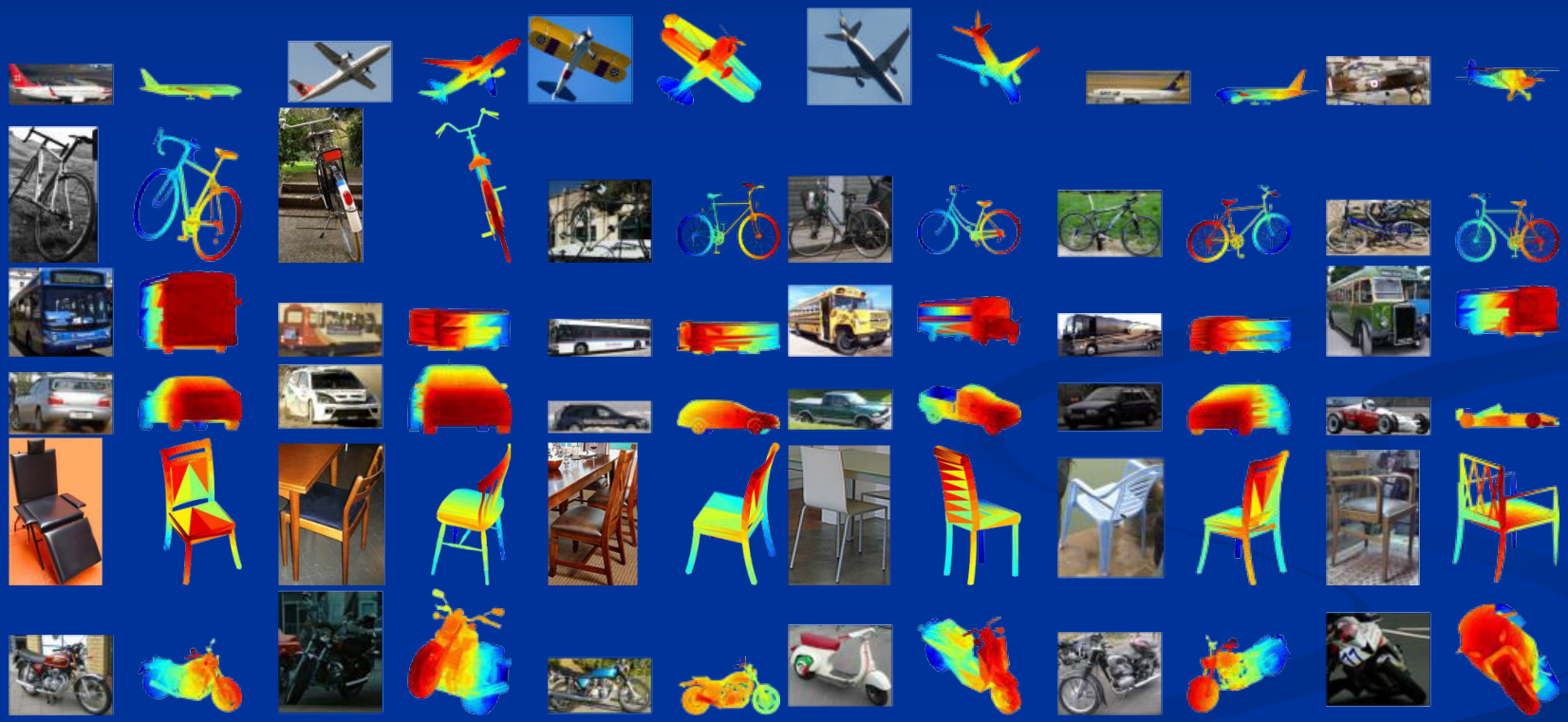
Viewpoint Prediction

- Viewpoint is described by elevation, azimuth and cyclorotation
- For each angle, the problem is treated as a classification among fixed bins

Keypoint Prediction : Viewpoint Based Prior

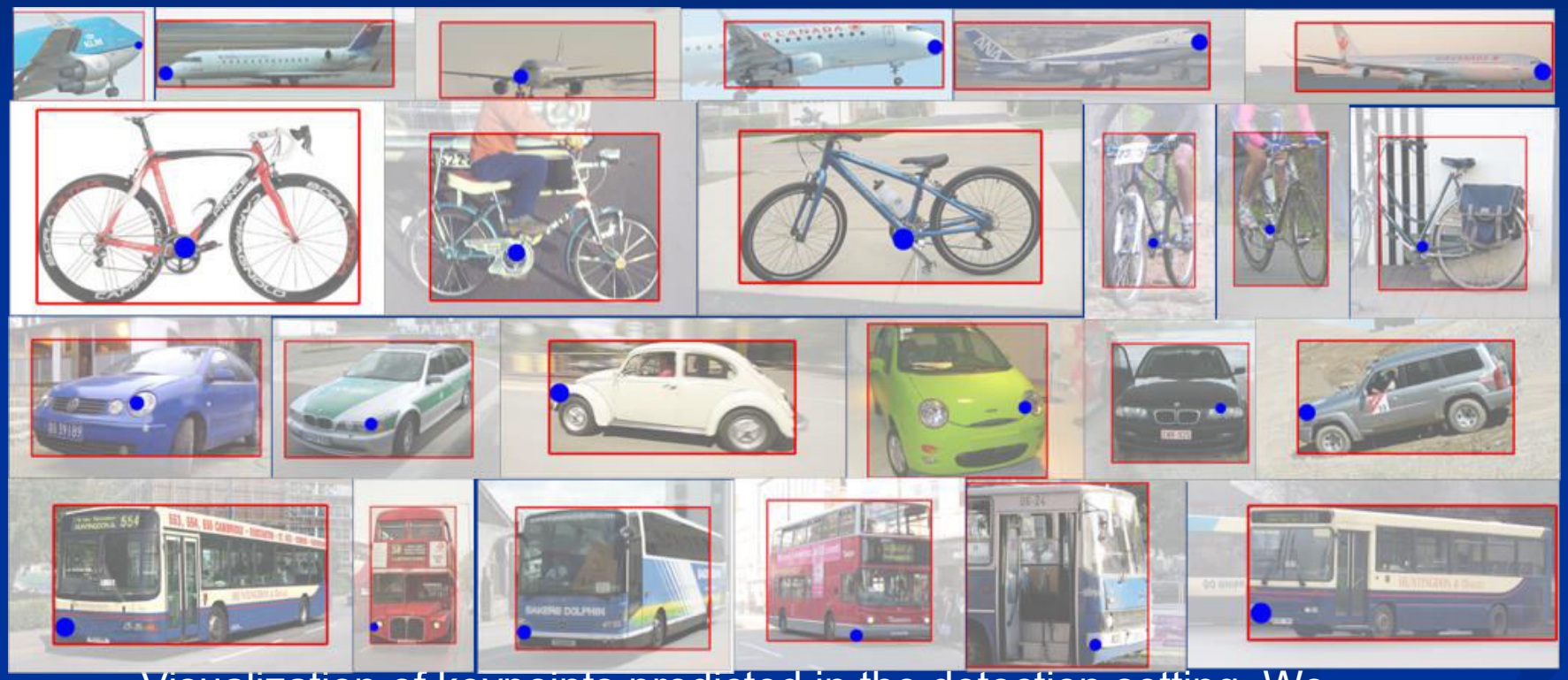
The viewpoint based location prior is computed using keypoint locations of training instances with similar views

Examples : Viewpoint Prediction



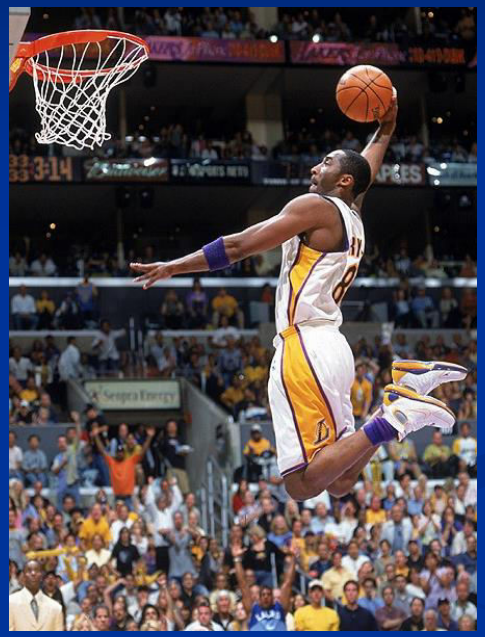
The columns show 15th, 30th, 45th, 60th, 75th and 90th percentile instances in terms of th

Examples : Keypoint Prediction



Visualization of keypoints predicted in the detection setting. We sort the keypoints detections by their prediction score and visualize every 15th detection for 'Nosetip' of aeroplanes, 'Left Headlight' of cars, 'Crankcentre' of bicycles and 'Left Base' of buses.

Action Recognition using Pose and Objects



Modeling Mutual Context of Object and Human Pose in Human-Object Interaction Activities, B. Yao and Li Fei-Fei, 2010

Slide Credit: Yao/Fei-Fei

Human-Object Interaction

Holistic image based classification

Integrated reasoning

- **Human pose estimation**

Slide Credit: Yao/Fei-Fei

Human-Object Interaction

Holistic image based classification

Integrated reasoning

- Human pose estimation
- **Object detection**

Slide Credit: Yao/Fei-Fei

Human-Object Interaction

Holistic image based classification

Integrated reasoning

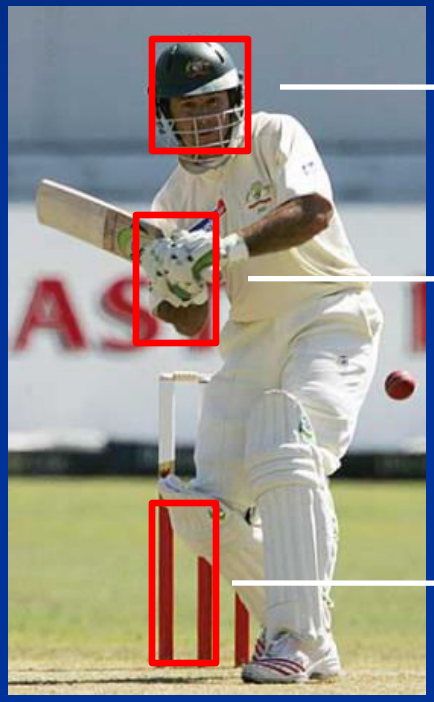
- **Human pose estimation**
- **Object detection**
- **Action categorization**

HOI activity: Tennis Forehand

Slide Credit: Yao/Fei-Fei

Human pose estimation & Object detection

Human pose estimation is challenging.



Difficult part appearance

Self-occlusion

Image region looks like a body part

- Felzenszwalb & Huttenlocher, 2005
- Ren et al, 2005
- Ramanan, 2006
- Ferrari et al, 2008
- Yang & Mori, 2008
- Andriluka et al, 2009
- Eichner & Ferrari, 2009

Slide Credit: Yao/Fei-Fei

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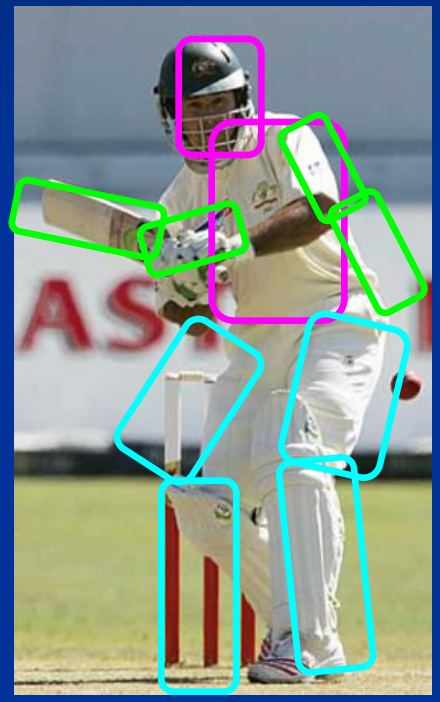

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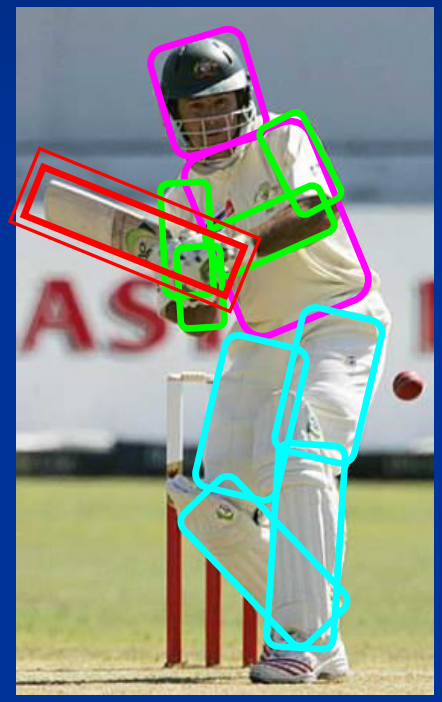
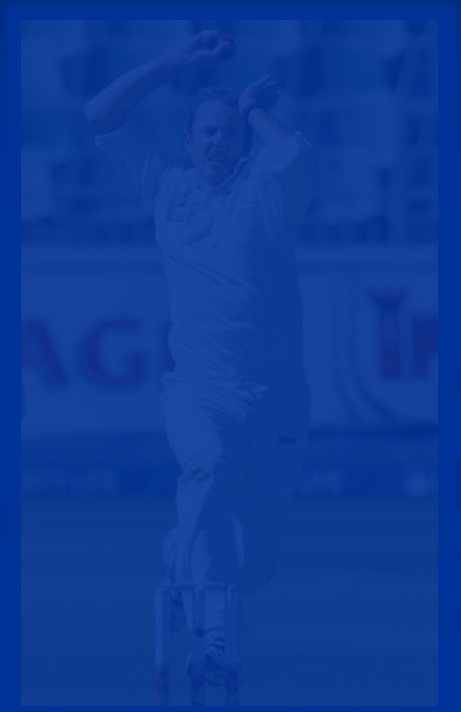
Slide Credit: Yao/Fei-Fei

Human pose estimation & Object detection

Facilitate

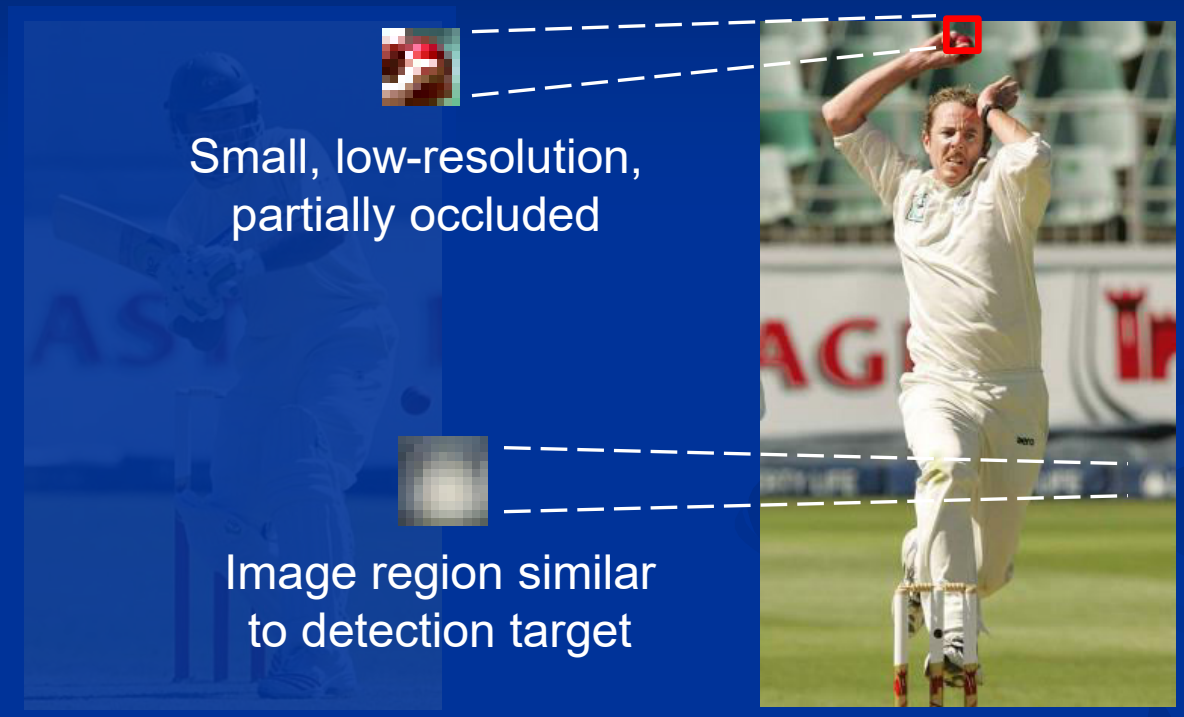
Given the object is detected.





Slide Credit: Yao/Fei-Fei

Human pose estimation & Object detection

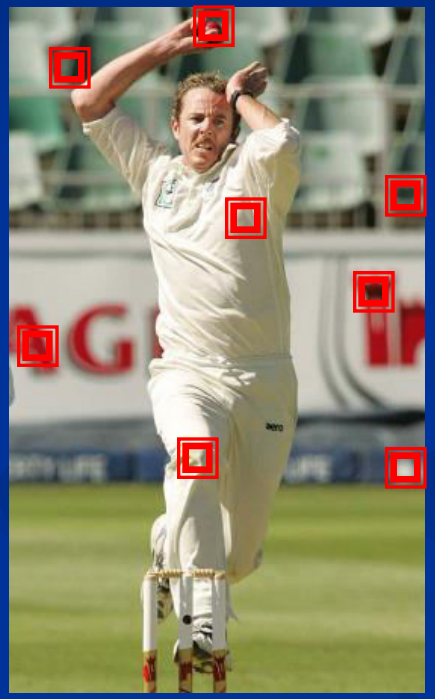


Object detection is challenging

- Viola & Jones, 2001
- Lampert et al, 2008
- Divvala et al, 2009
- Vedaldi et al, 2009

Slide Credit: Yao/Fei-Fei

Human pose estimation & Object detection



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- Viola & Jones, 2001
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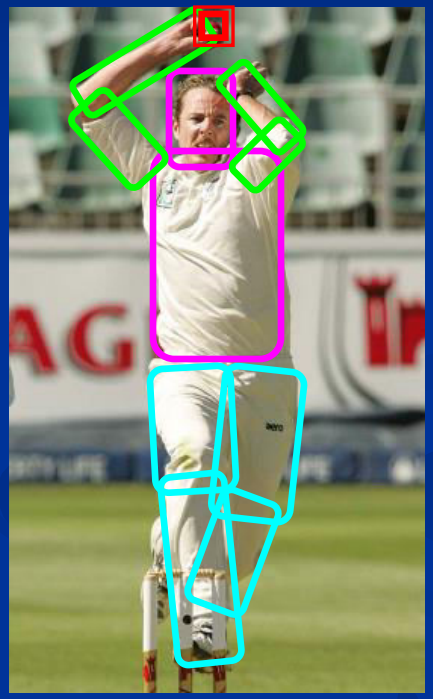
Human pose estimation & Object detection

Facilitate

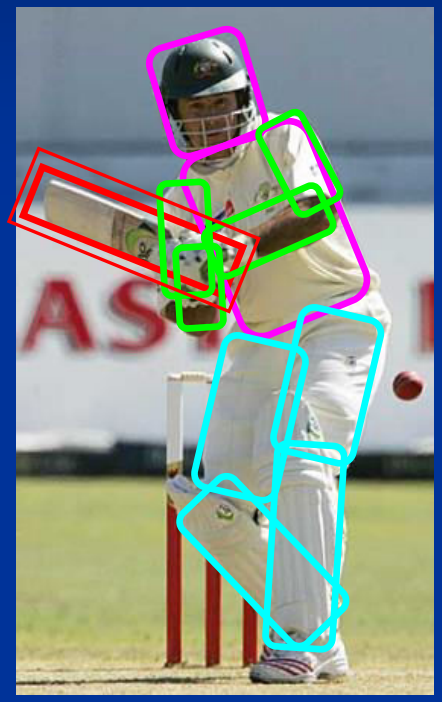


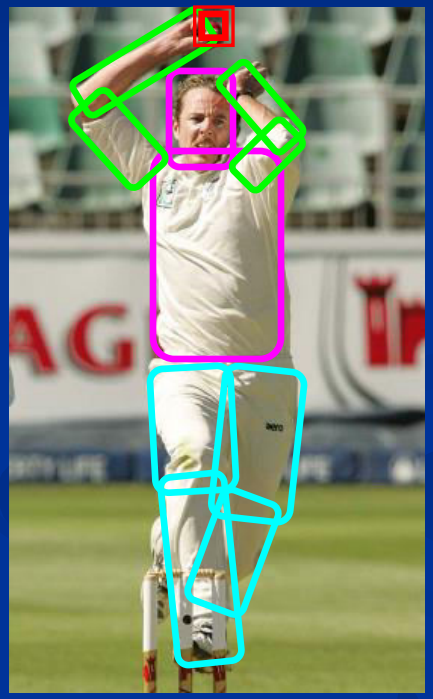
Given the pose is estimated.

Slide Credit: Yao/Fei-Fei

Human pose estimation & Object detection

Mutual Context





Slide Credit: Yao/Fei-Fei

Mutual Context Model Representation

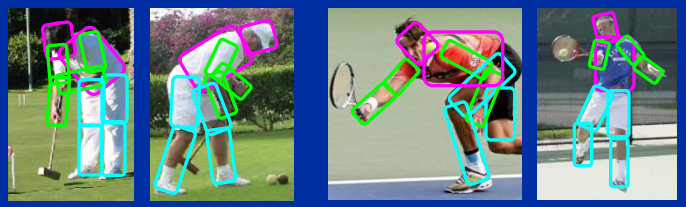
A:

Tennis forehand | Croquet shot | Volleyball smash | . . .

O:

Tennis racket | Croquet mallet | Volleyball | . . .

H:

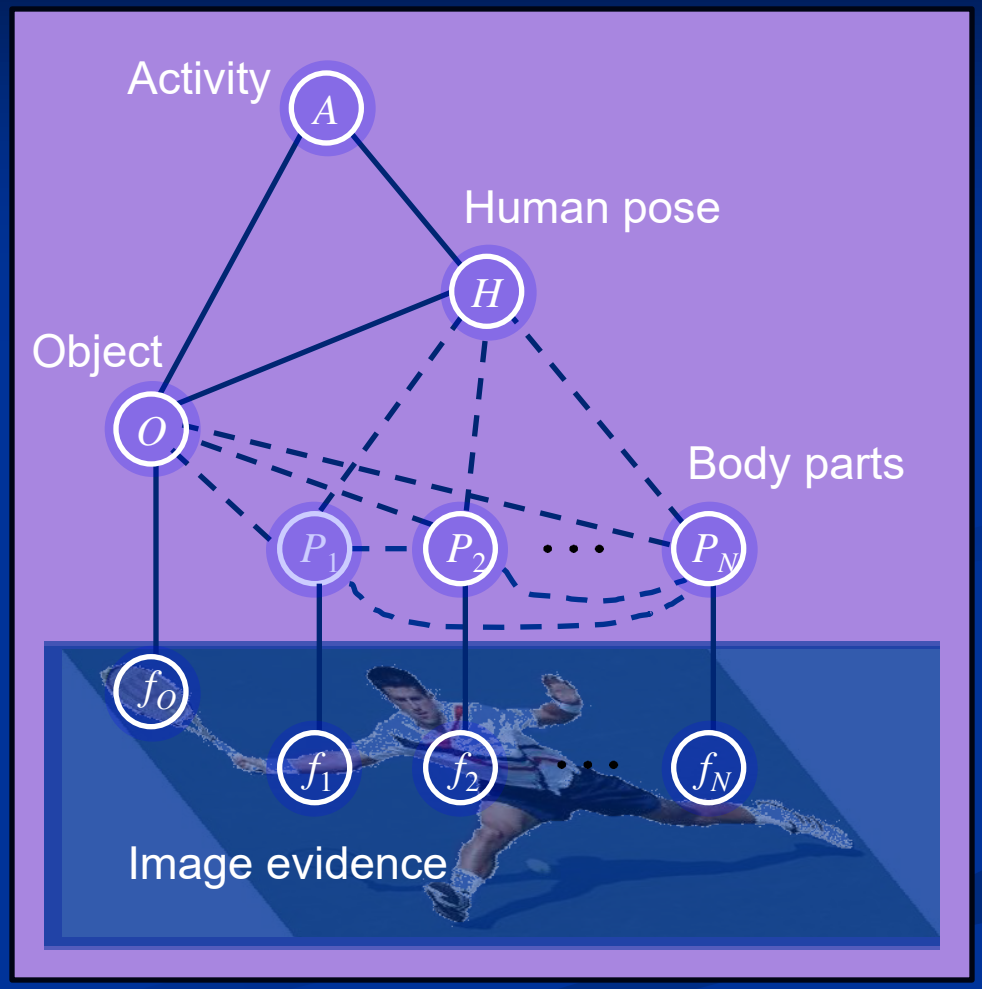


Intra-class variations

- More than one H for each A;
- **Unobserved** during training.

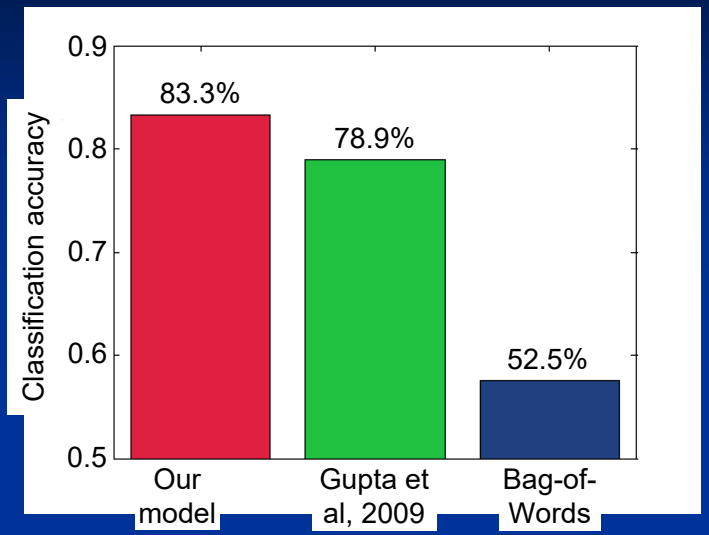
P: l_P: location; θ_P: orientation; s_P: scale.

f: Shape context. [Belongie et al, 2002]



Slide Credit: Yao/Fei-Fei

Activity Classification Results



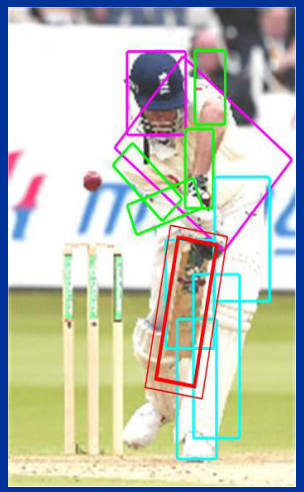
Yao and Li Fei-Fei, 2010

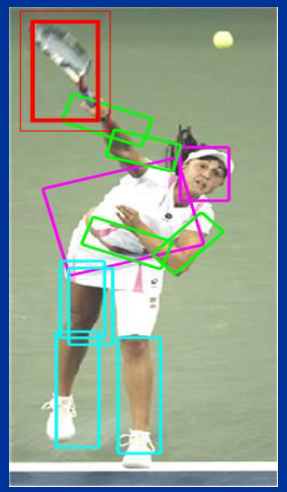
Gupta et al, 2009

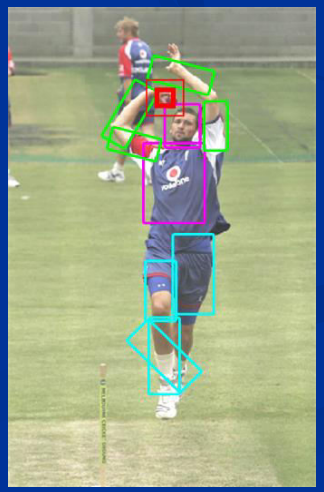
Bag-of-words SIFT+SVM

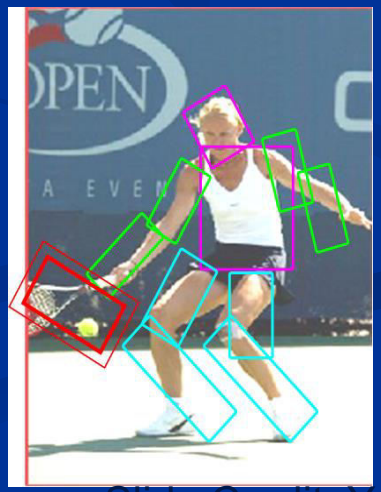
Cricket shot

Tennis forehand









Slide Credit: Yao/Fei-Fei

Other Examples

Joint Training of a Convolutional Network and a Graphical Model for Human Pose Estimation

Jonathan Tompson, Arjun Jain, Yann LeCun, Christoph Bregler
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Abstract

This paper proposes a new hybrid architecture that consists of a deep Convolutional Network and a Markov Random Field. We show how this architecture is successfully applied to the challenging problem of articulated human pose estimation in monocular images. The architecture can exploit structural domain constraints such as geometric relationships between body joint locations. We show that joint training of these two model paradigms improves performance and allows us to significantly outperform existing state-of-the-art techniques.

Key ideas

- Convolutional net based part detector predicts heat maps for keypoints.
- Use these as unary potentials for a MRF (Markov Random Field) with binary potentials that constrain joint inter-connectivity and global pose consistency (e.g. a face detector peak should not be very far from a shoulder detector peak)

3.1 Convolutional Network Part-Detector

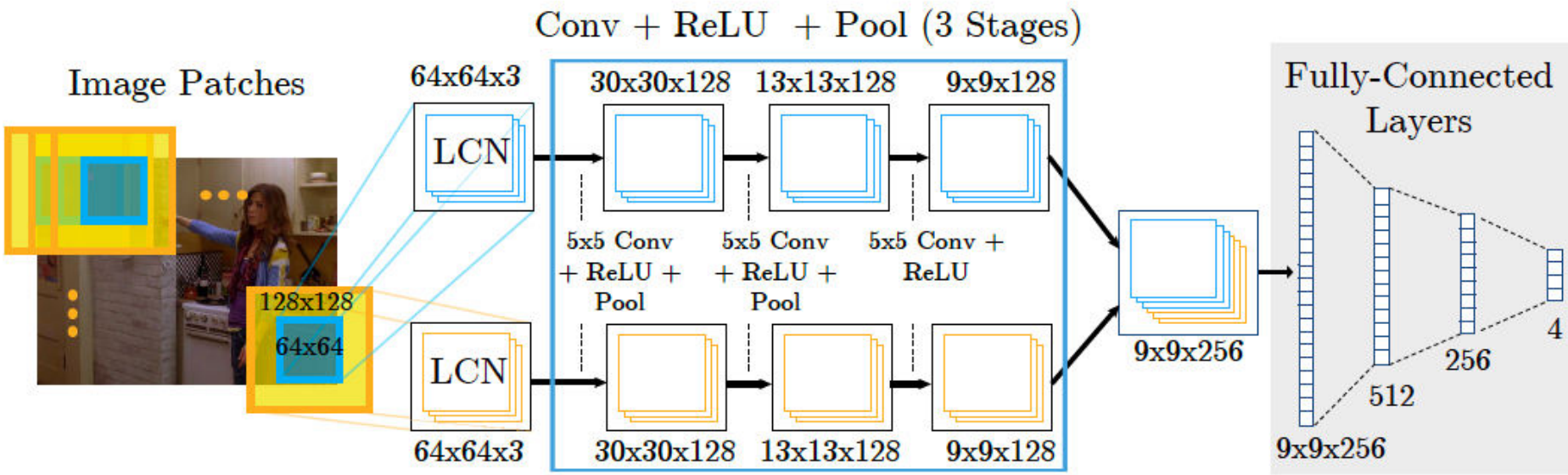

Figure 1: Multi-Resolution Sliding-Window With Overlapping Receptive Fields

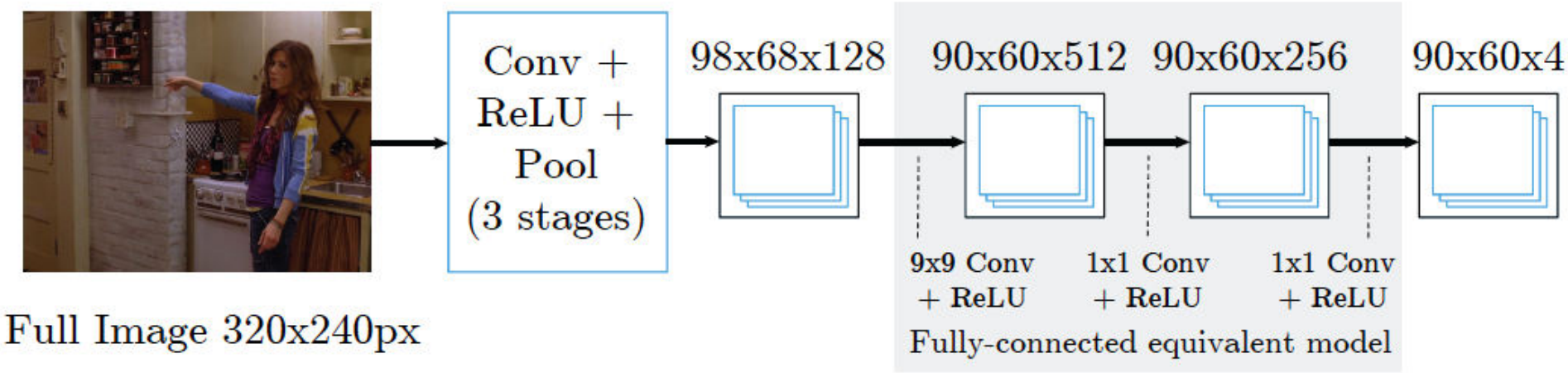

Figure 2: Efficient Sliding Window Model with Single Receptive Field

Convolutional priors

given that body part B is located at the center pixel, the convolution prior $P_{A|B}(i,j)$ is the likelihood of the body part A occurring in pixel location (i,j). For a body part A, we calculate the final marginal likelihood $\bar{p}_A$ as:

$$\bar{p}_A = \frac{1}{Z} \prod_{v \in V} \left(p_{A|v} * p_v + b_{v \to A} \right) \tag{1}$$

where v is the joint location, $p_{A|v}$ is the conditional prior described above, $b_{v \to a}$ is a bias term used to describe the background probability for the message from joint v to A

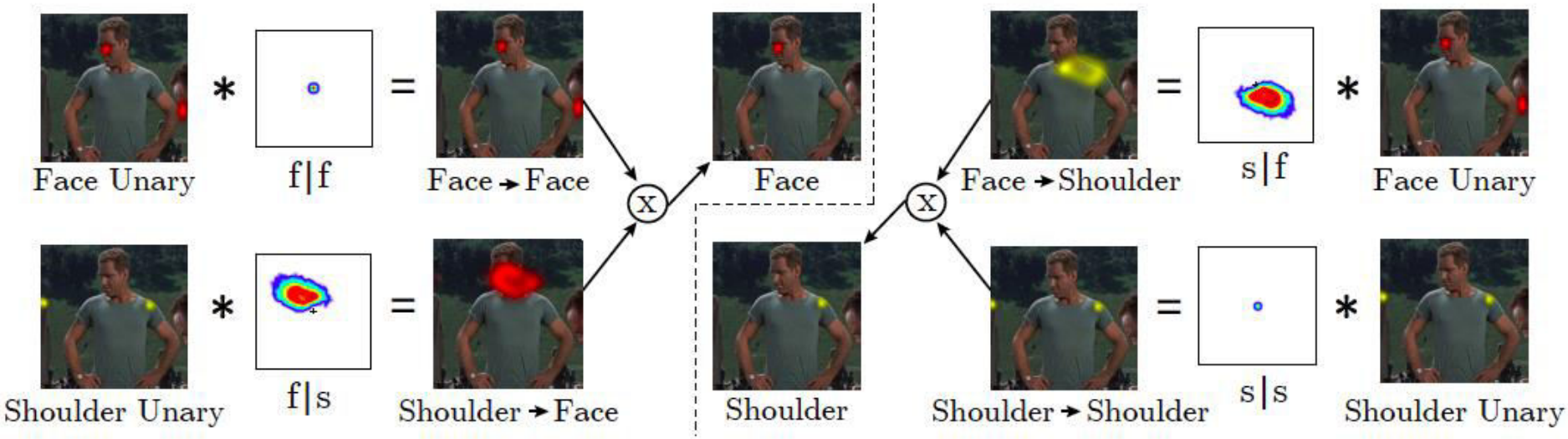

Figure 5: Didactic Example of Message Passing Between the Face and Shoulder Joints

International Journal of Computer Vision, 5:2, 195–212 (1990)
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Recognizing Solid Objects by Alignment with an Image

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Abstract
In this paper we consider the problem of recognizing solid objects from a single two-dimensional image of a three-dimensional scene. We develop a new method for computing a transformation from a three-dimensional model coordinate frame to the two-dimensional image coordinate frame, using three pairs of model and image points. We show that this transformation always exists for three noncollinear points, and is unique up to a reflective ambiguity. The solution method is closed-form and only involves second-order equations. We have implemented a recognition system that uses this transformation method to determine possible *alignments* of a model with an image. Each of these hypothesized matches is verified by comparing the entire edge contours of the aligned object with the image edges. Using the entire edge contours for verification, rather than a few local feature points, reduces the chance of finding false matches. The system has been tested on partly occluded objects in highly cluttered scenes.

Reconstruction of Articulated Objects from Point Correspondences in a Single Uncalibrated Image

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Received July 19, 1999; accepted September 11, 2000

This paper investigates the problem of recovering information about the configuration of an articulated object, such as a human figure, from point correspondences in a single image. Unlike previous approaches, the proposed reconstruction method does not assume that the imagery was acquired with a calibrated camera. An analysis is presented which demonstrates that there is a family of solutions to this reconstruction problem parameterized by a single variable. A simple and effective algorithm is proposed for recovering the entire set of solutions by considering the foreshortening of the segments of the model in the image. Results obtained by applying this algorithm to real images are presented. © 2000 Academic Press

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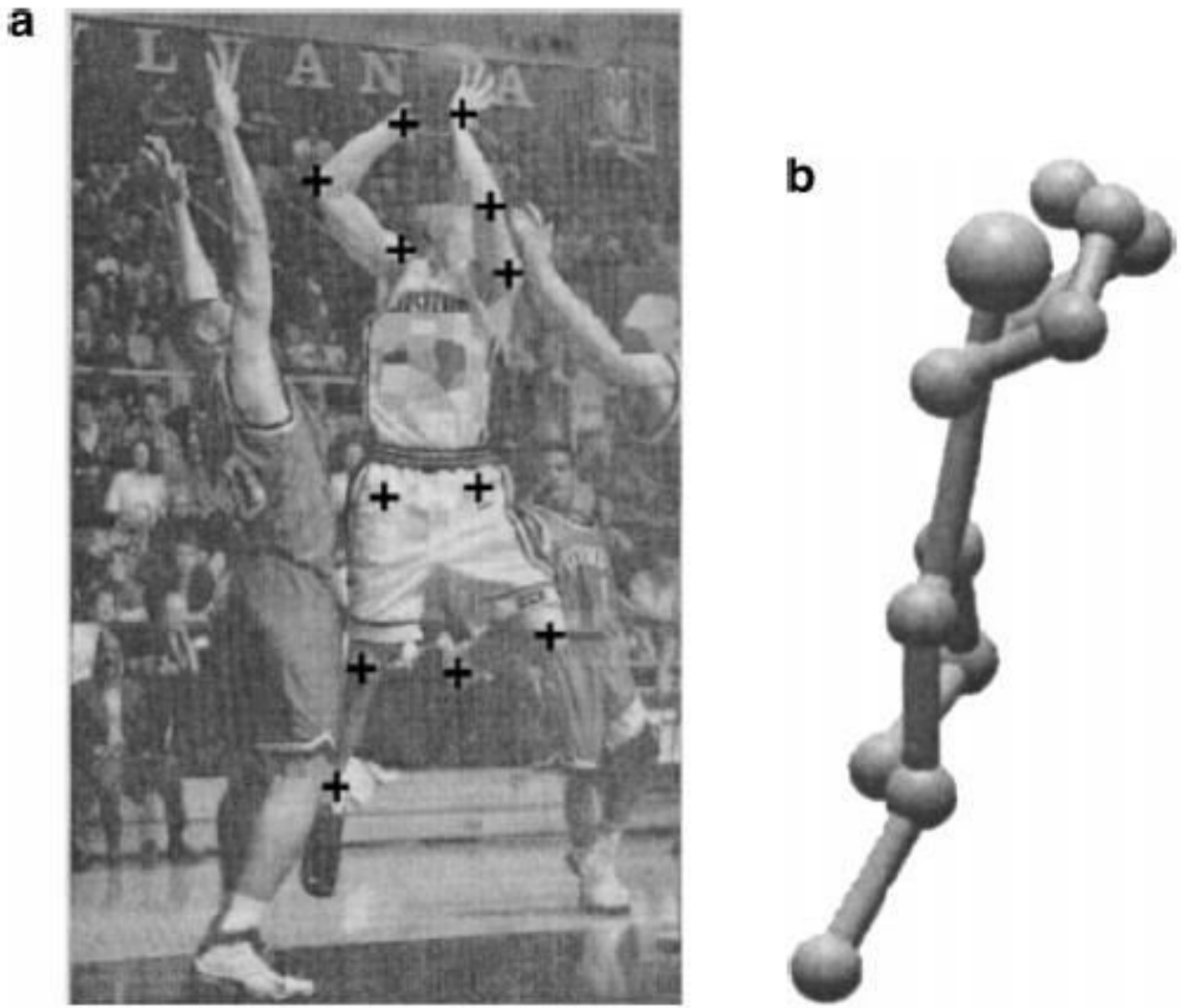

FIG. 1. (a) An image containing a figure to be recovered. The 12 crosses represent the estimated locations of the joints which are passed to the reconstruction procedure. (b) The recovered 3D model viewed from a novel vantage point.

Key observation: Lengths of various body segments are known